INCH-POUND
MIL-DTL-70683B(AR)
05 October 2021
SUPERSEDING
MIL-C-70683A(AR)
22 August 1991

DETAIL SPECIFICATION

CHARGE, DEMOLITION: 40LB, CRATERING PARTS FOR AND LOADING, ASSEMBLING AND PACKING

Reactivated after 05 October 2021 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers quality assurance provisions and preparation for delivery of the parts, loading, assembling and packing of Charge, Demolition: 40lb, Cratering.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM – Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1375

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-2440	- Container, Metal, For Propelling Charge
MIL-DTL-15011	- Pallets, Material Handling, Wood: Post Construction, 4-Way Entry
MIL-E-22267	- Explosive Compositions, HBX Type
MIL-DTL-50523	- Charge, Demolition, M112, Parts for Loading, Assembling and Packing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1916	- DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

CCDC – AC DRAWINGS

9395714	- Charge, Demolition: 40lb, Cratering
9395834	- Cover, Bottom
9395835	- Body
9395836	- Body Assembly
9395850	- Charge, Demolition: 40 lb., Cratering Loading Assembly
9395851	- Cover, Top
9395852	- Cover Top Assembly
12624777	- Packing and Marking for Container, Metal, Ammo, M18A2, for Charge, Demolition: 40lb, Cratering
12961084	- Combination of Adopted Items for Charge, Demolition: 40 lb, Cratering

(Application for copies of these drawings may be requested thru email to usarmy.pica.ccdc-ac.list.drawing-request-help-desk@mail.mil.)

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DEFENSE AMMUNITION CENTER (DAC)

ACV00561	-	Unit Load Marking for Shipment and Storage, Ammunition and Explosives
ACV00831	-	Heat Treatment of Wooden Pallets, Skids, Filler Assemblies, and Dunnage Assemblies
19-48-4079/11	-	Unitizing Procedures for Complete Rounds Packed in Cylindrical Metal Containers on 4-Way Entry Pallets: M18 Series Container

(Copies of these documents may be requested online at https://www.dau.edu/cop/ammo/Pages/Topics/Drawings.aspx.)

DEPARTMENT OF THE ARMY PUBLICATIONS

TM 3-34.82	-	Explosives and Demolitions
TM 9-1375-213-12	-	Operators and Unit Maintenance Manual (Including Repair Parts and Special Tools List): Demolition Materials

(Copies of publications may be obtained from https://armypubs.army.mil/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Materials and assemblies. All materials, parts, and assemblies shall comply with all requirements specified on Drawing (Dwg.) 9395714.

3.3 Strength of handle. The top cover assembly shall not be damaged by application of the specified static load to the handle.

3.4 Leak test. The body assembly shall not leak when submerged in water and pressurized in accordance with Dwg. 9395836.

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3.5 Waterproofness. The loaded assembly shall not leak when fully submerged with the top surface covered by not less than 0.5 inches and not greater than 3 inches of water and simultaneously subjected to a vacuum equivalent to 8 ± 1 inches of mercury for not less than 15 seconds.

3.6 Cast defects. Loaded assemblies shall be free of excessive cast defects in accordance with Dwg. 9395850.

3.7 Functioning. The loaded assembly shall function high-order when initiated by one (1) M112 Demolition Block conforming to MIL-DTL-50523.

3.8 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.9 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, weld splatter, sharp edges, chipped edges, gouges, cracks, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X	X	X		4.2
3.1.2	Conformance	X		X	X		X	4.3
3.2	Materials and assemblies	X		X	X	X	X	4.3.4.1 – 4.3.4.10
3.3	Strength of handle				X	X	X	4.4.1
3.4	Leak test				X	X	X	4.4.2
3.5	Waterproofness				X	X	X	4.4.3
3.6	X-ray			X		X	X	4.4.4
3.7	Functioning				X	X	X	4.4.5
3.8	Ammo lot numbering			X		X	X	4.3.4.7
3.9	Workmanship			X		X	X	4.3.4.1 – 4.3.4.8
5.1	Packaging			X	X	X	X	4.3.4.9 – 4.3.4.11
5.2	Pressure retention of ammo container				X	X	X	4.4.6
5.3	Unitization			X		X	X	4.3.4.11
5.4	Workmanship (packaging)			X		X	X	4.3.4.9 – 4.3.4.11

4.1 Classification of inspections. The following inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Submission. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table II. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE II. First article acceptance tests.

Examination or test	Sample quantity	Requirement paragraph	Inspection method
Cover, Bottom (Dwg. 9395834)			
Examination for defects	5	3.2	4.3.4.1
Body (Dwg. 9395835)			
Examination for defects	5	3.2	4.3.4.2
Body Assembly (Dwg. 9395836)			
Examination for defects	5	3.2	4.3.4.3
Leak Test	5	3.4	4.4.2
Cover, Top (Dwg. 9395851)			
Examination for defects	5	3.2	4.3.4.4
Cover, Top Assembly (Dwg. 9395852-1)			
Examination for defects	10	3.2	4.3.4.5
Strength of handle	10	3.3	4.4.1
Charge, Demolition: 40lb., Cratering Loading Assembly (Dwg. 9395850)			
Examination for defects	5	3.2	4.3.4.6
X-Ray	5	3.6	4.4.4
Charge, Demolition: 40lb., Cratering (Prior to Installation of Cover, Top Assembly (Dwg. 9395714-1)			
Examination for defects	5	3.2	4.3.4.7
Ammunition lot numbering	5	3.8	4.3.4.7
Charge, Demolition: 40 lb., Cratering (Dwg. 9395714-1)			
Examination for defects	20	3.2	4.3.4.8
Waterproofness	20	3.5	4.4.3
Functioning	20	3.7	4.4.5
Packaging (Dwg. 12624777)			
Examination for defects	5	5.1	4.3.4.9 – 4.3.4.10
Pressure retention test	5	5.2	4.4.6
Packaging (Dwg. 19-48-4079/11)			
Examination for defects	1 full pallet	5.3	4.3.4.110

4.2.2 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. Inspection lots shall contain materials from not more than one lot interfix number from one manufacturer (when applicable). When not applicable, lot formation shall be homogeneous. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance

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with the same drawings and revisions, and the same specifications and revisions. In addition each lot shall contain materials from not more than one lot interfix number from one manufacturer except the following:

a. Composition H-6 explosive charge mixture ingredients from not greater than one lot interfix from one manufacturer, of any ingredient.

4.3.2 Sampling. Unless otherwise noted, sampling of Major characteristics shall be in accordance with Verification Level IV and sampling of Minor characteristics shall be in accordance with Verification Level II of MIL-STD-1916.

4.3.3 Rejection. Unless otherwise noted, if any sample inspected in 4.3.4 or 4.4 fails to comply with any of the applicable requirements the lot shall be rejected.

4.3.4 Examination. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.11, the definitions of critical, major and minor defects are provided in paragraph "Definitions" of MIL-STD-1916. The acceptance criteria for sampling inspection shall be in accordance with the levels provided in conformance examination/test paragraph and MIL-STD-1916. Alternate conformance procedures may be proposed (See paragraph "Acceptance by contractor-proposed provisions" of MIL-STD-1916).

4.3.4.1 Cover, bottom.			Drawing Number 9395834 Next Higher Assembly 9395836	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major	None defined			
Minor				
201	Poor workmanship	VL-II	3.9	Visual

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4.3.4.2 Body.			Drawing Number 9395835 Next Higher Assembly 9395836	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Inside diameter	VL-IV	3.2	Gage
102	Evidence of full length weld	VL-IV	3.2	Visual
Minor				
201	Poor workmanship	VL-II	3.9	Visual

4.3.4.3 Body assembly.			Drawing Number 9395836 Next Higher Assembly 9395714-1	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Leak test	VL-IV	3.4	4.4.2
102	Outer diameter at crimped end, maximum (max.)	VL-IV	3.2	Gage
103	Height	VL-IV	3.2	Gage
104	Evidence of 360˚ weld	VL-IV	3.2	Visual
Minor				
201	Poor workmanship	VL-II	3.9	Visual

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4.3.4.4 Cover, top.			Drawing Number 9395851 Next Higher Assembly 9395852-1	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Presence sealing compound	VL-IV	3.2	Visual
Minor				
201	Poor workmanship	VL-II	3.9	Visual

4.3.4.5 Cover, top assembly.			Drawing Number 9395852-1 Next Higher Assembly 9395714-1	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Strength of handle	25	3.3	4.4.1
Minor				
201	Presence/proper orientation of handle, handle brackets and top cover	VL-II	3.9	Visual
202	Poor workmanship	VL-II	3.9	Visual

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4.3.4.6 Charge, demolition: 40 lb., cratering, loading assembly.			Drawing Number 9395850 Next Higher Assembly 9395714-1	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Weight of explosive	VL-IV	3.2	Gage
102	X-ray	25	3.6	4.4.4
103	Explosive composition	1/	3.2	2/
Minor				
201	Poor workmanship	VL-II	3.9	Visual
1/ Explosive composition shall be verified not less than three times per batch (see 6.7). Samples shall be collected from the beginning, middle, and end of every batch. No composite sampling shall be allowed. 2/ Inspection method shall be in accordance with MIL-E-22267.				

4.3.4.7 Charge, demolition: 40 lb., cratering (prior to installation of cover, top assembly.			Drawing Number 9395850 Next Higher Assembly 9395714-1	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Components missing	VL-IV	3.2	Visual
Minor				
201	Ammunition lot numbering	VL-II	3.8	Visual
202	Poor workmanship	VL-II	3.9	Visual

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4.3.4.8 Charge, demolition: 40 lb., cratering.			Drawing Number 9395714-1 Next Higher Assembly 12961084	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Waterproofness	100%	3.5	4.4.3
102	Functioning (1/)	5	3.7	4.4.5
103	Top cover assembly separates from body assembly when lifted by handle	5	3.2	Manual
Minor				
201	Poor workmanship	VL-II	3.9	Visual
202	Presence of explosive material on exterior of charge	VL-II	3.9	Visual
1/ After three consecutive lot acceptance tests of 5 samples are approved, the quantity shall be reduced to 3 samples. In event of a failure or first article test being performed, the quantity of lot acceptance test samples shall revert back to 5 until 3 consecutive lots meet the requirement specified.				

4.3.4.9 Container, metal, ammo, M18A2, for Charge, demo: 40 lb., cratering, prior to sealing.			Drawing Number 12624777 Next Higher Assembly -------	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major	None defined			
Minor				
201	Ammunition lot numbering	VL-II	3.8	Visual
202	Contents missing or incorrect	VL-II	5.1	Visual
203	Poor workmanship	VL-I	5.4	Visual

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4.3.4.10 Container, metal, ammo, M18A2, for Charge, demo: 40 lb., cratering.			Drawing Number 12624777 Next Higher Assembly 12961084	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Metallic seal missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL-II	5.1	Visual
102	Container punctured, dented, cracked, or damaged	VL-II	5.1	Visual
103	Contents of container exposed	VL-II	5.1	Visual
104	Pressure retention test	VL-II	5.2	4.4.6
Minor				
201	Contents loose	VL-I	5.1	Manual/visual
202	Container inadequately or improperly sealed	VL-I	5.1	Visual
203	Marking missing, incorrect, illegible, incorrect size/height, or not in proper location	VL-I	5.1	Visual
204	Poor workmanship	VL-I	5.4	Visual

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4.3.4.11 Unitized load for Charge, demo: 40 lb., cratering.			Drawing Number 19-48-4079/11 Next Higher Assembly 12961084	
Category	Examination or test	Conformance criteria	Requirement paragraph	Inspection method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL-II	5.3	Visual
102	Pallet load markings – including 2D bar codes – missing, incorrect, illegible, or not in proper location	VL-II	5.3	Visual
103	Manufacturer's mark not present	VL-II	5.3	Visual
104	Ammo stacked incorrectly	VL-II	5.1, 5.3	Visual
105	Incorrect number of charges	VL-II	5.1, 5.3	Visual
106	Marking incorrect	VL-II	5.1, 5.3	Visual
107	Banding applied incorrectly	VL-II	5.1, 5.4	Manual/visual
Minor				
201	Poor workmanship	VL-I	5.4	Visual

4.4 Test methods.

4.4.1 Strength of handle. The cover top assembly shall be assembled to the applicable test equipment (see 6.3) and not less than 100 pounds static load applied to the handle (see Dwg. 9395852-1). Observation shall be made for failure of a weld or handle.

4.4.2 Leak test. Body assembly (Dwg. 9395836) shall be submerged in water, pressurized and observation for leakage made. Leakage is defined as the presence during test of a stream, or streams, of bubbles of any frequency.

4.4.3 Waterproofness. The loaded assembly (Dwg. 9395714-1) shall be submerged in water, the required vacuum applied and observation for leakage made. Leakage is defined as the presence of a stream or streams of bubbles of any frequency. Any assembly exhibiting leakage shall be removed from the lot.

4.4.4 X-ray. Loaded assemblies shall be x-rayed in two planes, both in the transverse direction and orthogonal to each other. Resulting images shall be inspected for compliance with the requirements stated in Drawing 9395850.

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4.4.5 Functioning. All charges shall be tested above ground, observing all the safety requirements of TM 3-34.82. All charges are to be primed and initiated in accordance with applicable sections of TM 9-1375-213-12. Only one (1) M112 Demolition Block per charge shall be used. The M112 shall be taped or otherwise attached to the charge and initiated. Observation shall be made for complete high-order function and complete detonation of the charge.

4.4.6 Pressure retention of ammo container. The sealed M18A2 (Dwg. 12624777) shall be placed in an appropriate vessel, pressurized as required and observation for pressure retention made.

5. PACKAGING

5.1 Materials, components, and assemblies. The materials, components, and assemblies shall comply with all the requirements specified on the item's respective drawings and all drawings and specifications referenced thereon.

5.2 Pressure retention of ammo container. The sealed M18A2 ammunition container conforming to MIL-C-2440 shall not exhibit leakage when subjected to the conditions stated in Drawing 12624777.

5.3 Unitization.

5.3.1 Pallet. The pallet shall conform to MIL-DTL-15011; heat treat requirements for all non-manufactured wood used in the palletized load applies; and foreign manufacturers shall have the heat treatment of non-manufactured wood products verified in accordance with their National Plant Protection Organization's compliance program.

5.3.2 Unit load. The unit load shall conform to Drawing 19-48-4079/11; shall be securely strapped; and shall contain all subcomponents, subassemblies and markings conforming to requirements.

5.3.3 Marking. Each unit load shall be marked in accordance with ACV00561. All palletized load wood packaging marking shall be marked in accordance with ACV00831.

5.4 Workmanship (packaging). All parts, subassemblies, and assemblies shall be free of defects such as cracks and tears, and shall be clean by being free of dirt, grease, oil (except where specifically required), rust, salt deposits, corrosion, and other foreign matter. All packing shall be dry.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The 40 lb. Cratering Charge is primarily employed in cratering and ditching operations, but is also effective for destroying buildings, fortifications, and bridge abutments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Provisions for submission/resubmission of first article samples.
c. Requirement for ammunition data cards as specified in MIL-STD-1168.
d. Requirement for acceptance inspection equipment (AIE) designs.
e. Quantity required and delivery schedules.
f. Requirement for certificates of analysis on the Composition H-6 (see 6.6), and certificates of conformance on all remaining materials purchased by the contractor or subcontractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
g. Requirement for evidence that the explosives are manufactured by qualified sources within the last five years.

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to the Technical Agency (see 6.5).

6.4 Equivalent test method approval. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted to the Technical Agency (see 6.5). This description should include but not be limited to the procedures used, the accuracy and precision, and drawings of any special equipment required.

6.5 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: CCDC – AC, ATTN: FCDD-ACE-QMG, Picatinny Arsenal, NJ 07806-5000.

6.6 Consideration of data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, contract data requirements list, acceptance and description sheets (for explosives) are required for each lot. These sheets should be prepared and submitted by the contractor in accordance with MIL-STD-1171 or otherwise as directed by the contract.

6.7 Definitions.

6.7.1 Batch. A discrete quantity of material that has been manufactured through various unit operations to produce a substantially uniform final product.

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6.8 Subject term (key word) listing.

M18A2
Composition H-6
Aluminized Composition B

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army – AR

Preparing activity:
Army – AR
(Project 1375-2021-001)

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